Contest Problem Set 12125

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Number Sense 12125

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Name _____

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School _____

1. $212 + 21 =$ _____.
2. $517 + 296 =$ _____.
3. The hundreds digit of 12021 is _____.
4. $18 \times 20 =$ _____.
5. $7 + 7 \times 11 =$ _____.
6. The remainder of $65 \div 4$ is _____.
7. $29 + 51 + 43 + 48 =$ _____.
8. $443 - 189 =$ _____.
9. $65 \div 13 =$ _____.
10. (estimate) $429 + 715 + 286 =$ _____.
11. $11 \times 48 =$ _____.
12. The remainder of $188 \div 8$ is _____.
13. $524 - 317 - 108 =$ _____.
14. $19^2 =$ _____.
15. $45 \times 18 =$ _____.
16. *CDXCVI* in Arabic numerals is _____.
17. $33 \times 27 =$ _____.
18. $8 \times 17 \times 5 =$ _____.
19. $68 + 85 + 187 =$ _____.
20. (estimate) $401 \times 149 =$ _____.
21. The GCD of 20 and 35 is _____.
22. $93 \times 25 =$ _____.
23. The smaller of $\frac{7}{12}$ and $\frac{3}{5}$ is _____ (fraction).
24. $24 \times 15 \div 9 =$ _____.
25. If 3 teaspoons is equal to 1 tablespoon, then 27 tablespoons is equal to _____ teaspoons.
26. $31^2 =$ _____.
27. $756 \div 18 =$ _____.
28. $21 + 27 + 33 + 39 + 45 + 51 =$ _____.
29. $1 - \frac{1}{2} - \frac{1}{4} =$ _____ (fraction).
30. (estimate) $49 \times 79 \times 51 =$ _____.
31. $33\frac{1}{3}\% =$ _____ (fraction).
32. $75^2 =$ _____.
33. The perimeter of a square with area 49 is _____.
34. $31 \times 39 =$ _____.
35. $13 + 1.3 + 0.13 =$ _____ (decimal).
36. The LCM of 20 and 35 is _____.
37. $99 \times 73 =$ _____.
38. The number of multiples of 6 between 23 and 105 is _____.
39. 20% of 390 is _____.
40. (estimate) $\sqrt{656565} =$ _____.

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41. $101 \times 86 =$ _____.

42. The remainder of $1346 \div 11$ is _____.

43. $33^2 - 22^2 =$ _____.

44. The largest prime number less than 88 is ____.

45. $17 \times \frac{17}{18} =$ _____ (mixed number).

46. $11^3 =$ _____.

47. The side length of a regular hexagon with perimeter 96 is _____.

48. $89 \times 29 =$ _____.

49. The number of terms in the arithmetic sequence $9, 20, 31, \ldots, 174$ is _____.

50. (estimate) $833 \times 126 =$ _____.

51. $104 \times 122 =$ _____.

52. The diameter of a circle with area 144π is ____.

53. 302_4 in base 2 is _____ $_2$.

54. The sum of the terms of the arithmetic sequence $6, 11, 16, \ldots, 101$ is _____.

55. $\sqrt{6724} =$ _____.

56. $92 \times 92 =$ _____.

57. The median of the list $6, 4, 4, 7, 11, 8, 6$ is ____.

58. $222 \times 43 =$ _____.

59. $13\frac{1}{3}\%$ of 75 is _____.

60. (estimate) $428571 \times 9 =$ _____.

61. Two standard dice are rolled. The probability that the sum of the numbers shown is 7 or 11 is _____ (fraction).

62. $11\frac{7}{8} \times 11\frac{1}{8} =$ _____ (mixed number).

63. $0.7\overline{6} =$ _____ (fraction).

64. In a triangle, two angles each measure 74°, and the measure of the third angle is _____ $^\circ$.

65. $77 \times 125 =$ _____.

66. If $v = 24$, then $v^2 + 12v + 36 =$ _____.

67. $7 \times 5 \times 14 \times 19 =$ _____.

68. The number 88 written in base 4 is _____ $_4$.

69. The perimeter of an equilateral triangle with area $\sqrt{3}$ is _____.

70. (estimate) $7^4 \cdot 5^4 =$ _____.

71. $1003 \times 1054 =$ _____.

72. The number of positive whole number divisors of 63 is _____.

73. $\sqrt{75} \times \sqrt{108} =$ _____.

74. The reciprocal of $\frac{25}{63}$ is _____ (decimal).

75. The greatest possible perimeter of a triangle with whole number side lengths and two sides of length 8 and 15 is _____.

76. $295^2 =$ _____.

77. $\frac{2}{7}$ of 44 is $\frac{11}{14}$ of _____.

78. $28 \times 19 \times 75 =$ _____.

79. The sum of the prime divisors of 385 is ____.

80. (estimate) $1.2 \times 2.2 \times 2.8 \times 1.8 =$ _____.

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Select E only if you cannot determine a uniquely correct answer between A, B, C, and D.

1.	A	B	C	D	E	11.	A	B	C	D	E	21.	A	B	C	D	E
2.	A	B	C	D	E	12.	A	B	C	D	E	22.	A	B	C	D	E
3.	A	B	C	D	E	13.	A	B	C	D	E	23.	A	B	C	D	E
4.	A	B	C	D	E	14.	A	B	C	D	E	24.	A	B	C	D	E
5.	A	B	C	D	E	15.	A	B	C	D	E	25.	A	B	C	D	E
6.	A	B	C	D	E	16.	A	B	C	D	E	26.	A	B	C	D	E
7.	A	B	C	D	E	17.	A	B	C	D	E	27.	A	B	C	D	E
8.	A	B	C	D	E	18.	A	B	C	D	E	28.	A	B	C	D	E
9.	A	B	C	D	E	19.	A	B	C	D	E	29.	A	B	C	D	E
10.	A	B	C	D	E	20.	A	B	C	D	E	30.	A	B	C	D	E

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1. Molly calculated the product of 20 and 21. What is the sum of the digits of Molly's result?

(A) 2 (B) 8 (C) 4 (D) 6 (E) Other

2. In the sport of basketball, some shots are worth 2 points, and some shots are worth 3 points. In a game, Allen made six shots worth 2 points and three shots worth 3 points. He did not score any other points in the game. How many points did Allen score in the game?

(A) 21 (B) 18 (C) 24 (D) 19 (E) Other

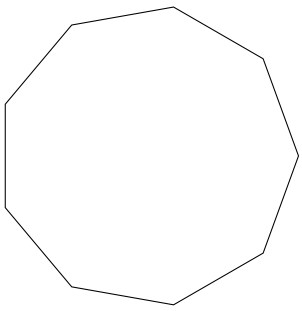
3. What is the sum of all of the single-digit positive whole numbers that evenly divide 72?

(A) 24 (B) 32 (C) 33 (D) 25 (E) Other

4. Somer gave the bank employee one $50 bill. The bank employee gave the same amount of money back to Somer using different bills. The bank employee gave Somer five $1 bills, two $10 bills, and the remainder of the bills were $5 bills. How many $5 bills did Somer receive from the bank employee?

(A) 6 (B) 7 (C) 3 (D) 5 (E) Other

5. A nonagon is a polygon with nine sides. In a regular nonagon, all of the sides have the same length. A regular nonagon has a perimeter of 171. What is the side length of the nonagon?



(A) 23 (B) 19 (C) 15 (D) 21 (E) Other

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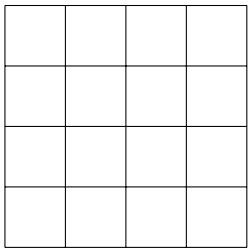
6. In the country Swapistan, there are two coins, the Duo and the Trio. Three Duos is worth the same amount as two Trios. Steve has thirty-six Trios. If he converts all of his Trios to Duos, then how many Duos will Steve have?

(A) 24 (B) 54 (C) 60 (D) 48 (E) Other

7. A prime number is a number that is only evenly divisible by itself and 1. The first few prime numbers are 2, 3, 5, and 7. Samantha summed all of the numbers that were greater than 11 and less than 20 that are not prime. What was Samantha's sum?

(A) 60 (B) 67 (C) 61 (D) 75 (E) Other

8. A square piece of paper eight inches on a side is cut into identical squares by six straight cuts, as shown below. Eileen puts nine of the smaller squares into the recycle bin. What is the total perimeter of the remaining squares?



(A) 72 (B) 48 (C) 36 (D) 28 (E) Other

9. Alice, Bobby, Carrie, and Denise need to line up for a photo. How many different ways can they line up if Alice must be next to Carrie, but Denise cannot be next to Bobby?

(A) 1 (B) 2 (C) 4 (D) 6 (E) Other

10. What is the sum of the positive whole numbers less than 160 that can be evenly divided by both 3 and 5?

(A) 675 (B) 750 (C) 990 (D) 825 (E) Other

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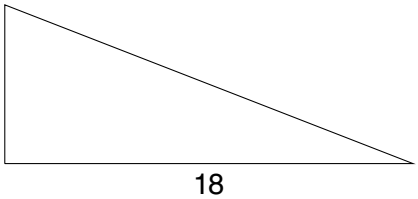
11. At her family's holiday dinner, Cecily cut the pie. She cut five pieces that were each $8\frac{1}{3}\%$ of the pie, and two pieces that were each 20% of the pie. The final piece was the remainder of the pie. What fraction of the pie was the final piece?

(A) $\frac{13}{60}$ (B) $\frac{1}{5}$ (C) $\frac{11}{60}$ (D) $\frac{7}{30}$ (E) Other

12. What is the value of $\left(\left(\left(\left(1+\frac{1}{2}\right)\div\left(1+\frac{1}{3}\right)\right)\div\left(1+\frac{1}{4}\right)\right)\div\left(1+\frac{1}{5}\right)\right)$?

(A) $\frac{1}{5}$ (B) 1 (C) $\frac{4}{5}$ (D) $\frac{3}{4}$ (E) Other

13. A right triangle has an area of 63. One leg of the right triangle has length 18. What is the least positive whole number that is greater than the length of the hypotenuse?



(A) 19 (B) 22 (C) 21 (D) 20 (E) Other

14. There are twenty-three students in Mrs. Norman's fifth grade class. Five of the students are not in either art or Spanish. Three students are in both art and Spanish. The total number of students in art is twice the total number of students in Spanish. How many students are in art, but not in Spanish?

(A) 8 (B) 7 (C) 4 (D) 11 (E) Other

15. Prize tokens at the arcade are worth either five points or two points. Pierre has tokens worth a total of twenty-seven points. What is the sum of all possible values of the number of tokens that Pierre could have?

(A) 6 (B) 15 (C) 18 (D) 27 (E) Other

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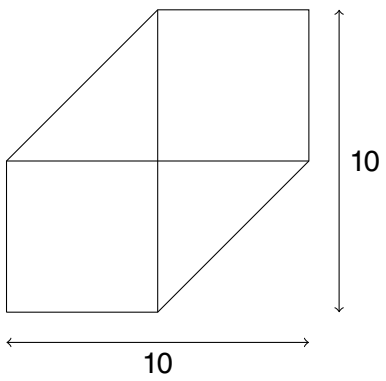
16. A square piece of paper measures 10 inches on a side. Opposite corners of the piece of paper are folded until they meet at the center of the square, and the folded over portions are then cut off and removed. To the nearest whole number, in inches, what is the perimeter of the resulting piece of paper?



(A) 40 (B) 36 (C) 32 (D) 38 (E) Other

17. Jamele chose a positive whole number and then divided it by 41. Her result was greater than $\frac{1}{5}$, but less than $\frac{1}{2}$. How many different positive whole numbers could Jamele have chosen?

(A) 12 (B) 11 (C) 10 (D) 13 (E) Other

18. How many even positive whole numbers evenly divide 1728?

(A) 18 (B) 22 (C) 20 (D) 24 (E) Other

19. Allen formed a list of numbers by starting with the number 2. He then created additional numbers in the list by alternating between subtracting 1 from the previous number and adding 5 to the previous number. The first few numbers in Allen's list were 2, 1, 6, 5, 10, 9, and 14. What is the sum of the first forty numbers in Allen's list?

(A) 1600 (B) 1610 (C) 1640 (D) 1580 (E) Other

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20. If $m\spadesuit n = m^3 - 3m^2n + 3mn^2 - n^3$, then what is the value of $(3\spadesuit 1)\spadesuit(2\spadesuit 1)$?

(A) 360 (B) 343 (C) 350 (D) 324 (E) Other

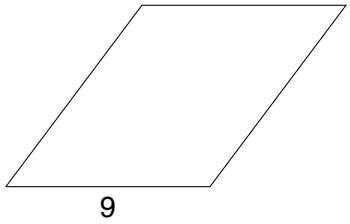
21. At Igloo Homemade Ice Creams, customers can customize their sundae by choosing any two of the five available flavors, and then some, all, or none of the three available toppings. How many different sundaes are possible at Igloo Homemade Ice Creams?

(A) 72 (B) 96 (C) 80 (D) 84 (E) Other

22. If $3^{2x+1} = 54$, then what is the value of 9^{x-1}?

(A) 6 (B) 18 (C) 9 (D) 4 (E) Other

23. A parallelogram has a side of length 9 and an area of 72. The longer diagonal of the parallelogram has length 17. What is the perimeter of the parallelogram?



(A) 38 (B) 44 (C) 42 (D) 46 (E) Other

24. What is the greatest power of 6 that evenly divides 120!?

(A) 6^{58} (B) 6^{60} (C) 6^{54} (D) 6^{55} (E) Other

25. Frank flips a fair coin five times. What is the probability that the same result, either heads or tails, does not occur on three or more consecutive flips?

(A) $\frac{7}{18}$ (B) $\frac{4}{9}$ (C) $\frac{5}{9}$ (D) $\frac{1}{2}$ (E) Other

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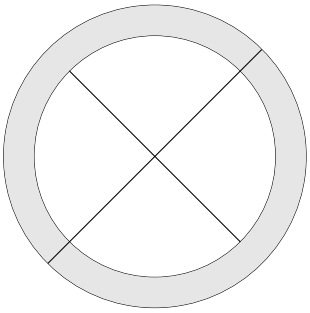
26. Jamie, Kevin, and Lara each chose a two-digit positive whole number with a units digit of 3. The product of their three numbers had a tens digit of 3. What is the tens digit of the sum of their three numbers?

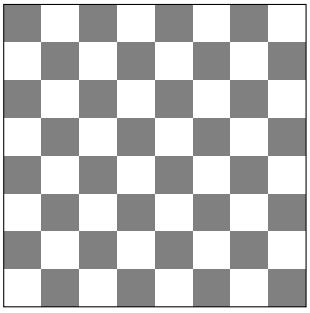
(A) 7 (B) 9 (C) 5 (D) 1 (E) Other

27. Two circles with the same center have diameters that are positive whole number lengths. The area of the region that lies within exactly one of the circles is 36π. What is the sum of all possible values of the diameter of the smaller circle?



(A) 56 (B) 30 (C) 65 (D) 25 (E) Other

28. An 8×8 checkerboard is shown below. A number of rectangles exist on this board, with boundaries of each rectangle that are boundaries of the small squares or boundaries of the board. A single small square is one such rectangle, and the entire board is another such rectangle. One of these rectangles is randomly chosen. The probability that it is a square can be expressed as a common fraction. What is the sum of the numerator and denominator of that fraction?



(A) 118 (B) 108 (C) 125 (D) 110 (E) Other

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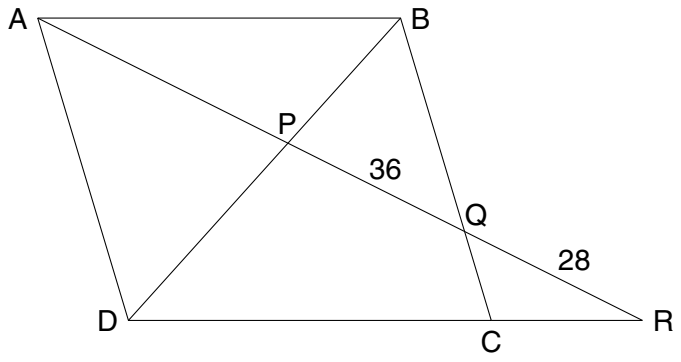
29. How many positive whole numbers less than or equal to 999 require six or fewer uses of the digit 1 when written in binary, or base 2?

(A) 845 (B) 832 (C) 841 (D) 847 (E) Other

30. In parallelogram *ABCD*, side *DC* is extended beyond *C* to point *R*. Segment *AR* intersects segment *BC* at point *Q* and intersects segment *BD* at point *P*. The length of segment *PQ* is 36 and the length of segment *QR* is 28. What is the length of segment *AP*?



(A) 42 (B) 45 (C) 48 (D) 44 (E) Other

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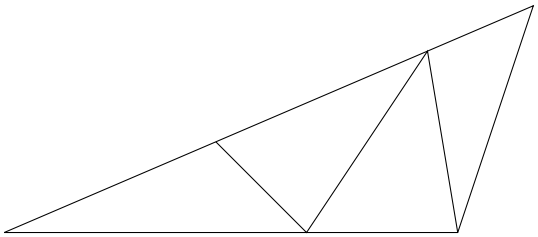
Problems 1 & 2

1. Trey chose a positive whole number greater than 4. He then summed all of the positive whole numbers from 5 through his number. For example, if the number Trey chose was 9, then his sum would be $5 + 6 + 7 + 8 + 9 = 35$. The value of Trey's sum was 200. What number did Trey choose?

1.

2. Using only line segments shown for sides of the triangle, how many triangles are in the figure below?

2.



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Problems 3 & 4

3. A perfect square is the result when a whole number is multiplied by itself. For example, since $7 \times 7 = 49$, the number 49 is a perfect square. How many positive whole numbers less than 1000 are perfect squares and have a units digit that is odd?

3.

4. A 500 gallon water tank is 40% full. At noon, Valerie opens a valve that begins to fill the tank at a rate of $2\frac{1}{2}$ gallons of water per minute. Sixteen minutes later, the tank springs a leak that allows water to leave the tank at a rate of $\frac{1}{3}$ gallon per minute. How many minutes after noon will the tank finally be full?

4.

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Problems 5 & 6

5. Dimes are coins worth ten cents and nickels are coins worth five cents. Dylan has two more dimes than he has nickels, and the total value of Dylan's nickels and dimes is 155 cents. What is the total number of nickels and dimes that Dylan has?

5.

6. Three standard six-sided dice are rolled. What is the probability that the product of the numbers shown is a multiple of four? Express your answer as a common fraction.

6.

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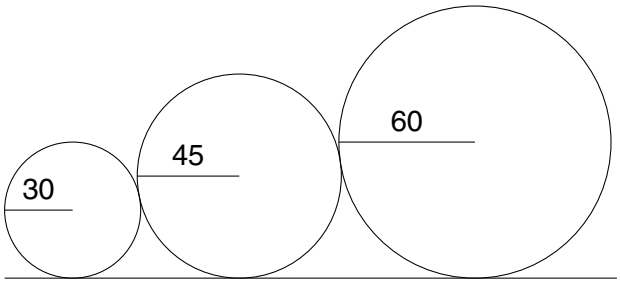
Problems 7 & 8

7. January 1st, 2021 is a Friday. In the year 2021, the month of January has 31 days, the month of February has 28 days, the month of March has 31 days, and the month of April has 30 days. What is the total number of Fridays and Saturdays in the month of April, 2021?

7.

8. A circle of radius 30 is externally tangent to a circle of radius 45. That same circle of radius 45 is externally tangent to a circle of radius 60. Additionally, a line is tangent to all three circles. To the nearest whole number, what is the area of the triangle with vertices at the centers of the three circles?

8.



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Team Round
12125

School or Team

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1.	2.	3.	4.	5.
6.	7.	8.	9.	10.

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1. Seth started watching the most recent episode of The Mandalorian at 12:36 pm. The episode finished at 1:27 pm. How many minutes long was the episode?

2. Ralph chose a two-digit positive whole number. When Ralph reversed the digits of his whole number, the new number was 45 more than the original number. How many different numbers could Ralph have chosen?

3. The average age of the six people on a winter ski trip was twenty-six. An eleven year-old had to leave the trip early to attend school, and a thirty-seven year-old left as well to take the eleven year-old home. What was the average age of the remaining people on the ski trip?

4. Steph and Beth live six miles away from each other. Steph leaves her house walking toward Beth's house at a speed of $3\frac{1}{2}$ miles per hour. Beth leaves her house for Steph's house at the same time, but she is riding her bike at a speed that is four times as fast as Steph walks. When Steph and Beth meet, how many miles more has Beth biked than Steph walked? Express your answer as a mixed number.

5. The magic square shown below is a 3×3 grid of smaller squares. The sum of the three numbers in any row, column, and diagonal of the grid is the same. What is the sum of the numbers that belong in the five empty smaller squares?

	17	
21		
22		19

6. The complement of an angle is the angle that produces a result of 90° when summed with the given angle. For example, since $31^\circ + 59^\circ = 90^\circ$, the complement of 31° is 59°, and the complement of 59° is 31°. Angela measured an angle with a positive degree measure less than 90°. The measure of Angela's angle was 9° less than half its complement. What was the measure of Angela's angle, in degrees?

7. Mia, Nala, and Omar each chose a positive whole number greater than one. The three whole numbers they chose may not have all been different. The product of the three numbers they chose was 72. How many different selections of numbers could Mia, Nala, and Omar have made?

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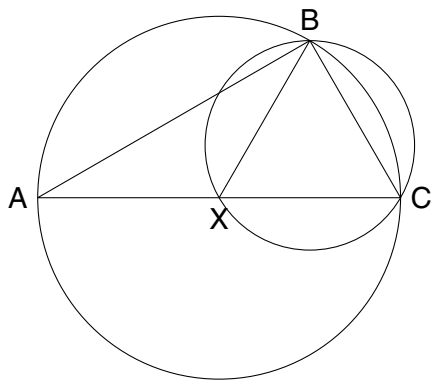
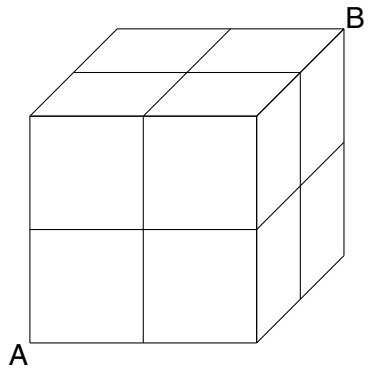
8. As a fundraiser, the Pineville Elementary Math Team sold raffle tickets. Some of the tickets were sold for full price, and some of the tickets were sold for half price. Both the full and half price tickets were priced at a whole number of dollars. The Pineville Elementary Math Team sold a total of 80 raffle tickets, and received a total of \$1122 from the sale of the raffle tickets. How many half-price raffle tickets did the Pineville Elementary Math Team sell?

9. In the figure below, point X lies on line segment AC, and a large circle with center at X is circumscribed about triangle ABC. A small circle is circumscribed about equilateral triangle BCX. What is the ratio of the area of the small circle to the area of the large circle? Express your answer as a common fraction.



10. A $2 \times 2 \times 2$ cube is formed by eight smaller $1 \times 1 \times 1$ cubes, as shown. An ant moves from corner A to corner B, staying on the exterior of the large cube, including the bottom of the cube, and traveling only along the edges of the large cube and the seams between the smaller cubes. Every move the ant makes on its journey moves the ant closer to B than to A. How many different paths can the ant take from corner A to corner B?



Answers 12125

Sprint Round

1. D
2. A
3. C
4. D
5. B
6. B
7. D
8. E
9. C
10. D
11. C
12. D
13. D
14. D
15. D
16. E
17. A
18. D
19. D
20. B
21. C
22. E
23. A
24. A
25. D
26. B
27. C
28. C
29. A
30. C

Target Round

1. 20
2. 7
3. 16
4. 136
5. 20
6. $\frac{5}{8}$
7. 9
8. 228

Team Round

1. 51
2. 4
3. 27
4. $3\frac{3}{5}$
5. 137
6. 24(°)
7. 27
8. 58
9. $\frac{1}{3}$
10. 54

Number Sense

1. 233
2. 813
3. 0
4. 360
5. 84
6. 1
7. 171
8. 254
9. 5
10. $[1359, 1501]$
11. 528
12. 4
13. 99
14. 361
15. 810
16. 496
17. 891
18. 680
19. 340
20. $[56762, 62736]$
21. 5
22. 2325
23. $\frac{7}{12}$
24. 40
25. 81
26. 961
27. 42
28. 216
29. $\frac{1}{4}$
30. $[187550, 207292]$
31. $\frac{1}{3}$
32. 5625
33. 28
34. 1209
35. 14.43
36. 140
37. 7227
38. 14
39. 78
40. $[770, 850]$
41. 8686
42. 4
43. 605
44. 83
45. $16\frac{1}{18}$
46. 1331
47. 16
48. 2581
49. 16
50. $[99711, 110205]$
51. 12688
52. 24
53. 110010
54. 1070
55. 82
56. 8464
57. 6
58. 9546
59. 10
60. $[3664283, 4049995]$
61. $\frac{2}{9}$
62. $132\frac{7}{64}$
63. $\frac{23}{30}$
64. 32
65. 9625
66. 900
67. 9310
68. 1120
69. 6
70. $[1425594, 1575656]$
71. 1057162
72. 6
73. 90
74. 2.52
75. 45
76. 87025
77. 16
78. 39900
79. 23
80. $[13, 13]$

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Solutions 12125

Sprint Round Solutions

1. The product is 20×21, which is $21 \times 2 \times 10$, or $42 \times 10 = 420$. The sum of the digits is $4 + 2 + 0 = \boxed{6}$.

2. Allen scored $6 \times 2 = 12$ points on shots worth 2 points, and $3 \times 3 = 9$ points on 3 shots worth 3 points, for a total number of points of $12 + 9 = \boxed{21}$.

3. The single-digit positive whole numbers that evenly divide 72 are 1, 2, 3, 4, 6, 8, and 9, and their sum is $1 + 2 + 3 + 4 + 6 + 8 + 9 = \boxed{33}$.

4. The total amount of money already received must be \$50. Already, Somer has received $5 \times \$1 = \$5$ and $2 \times \$10 = \$20$, for a total of \$25. The value of the \$5 bills must be $\$50 - \$25 = \$25$. The number of \$5 bills Somer received is $\$25 \div \$5 = \boxed{5}$.

5. Since all of the sides have the same length, the side length of the nonagon is $171 \div 9 = \boxed{19}$.

6. Steve's 36 Trios can be thought of as 18 sets of 2 Trios. This is equivalent to 18 sets of 3 Duos, so the equivalent number of Duos is $18 \times 3 = \boxed{54}$.

7. The prime numbers greater than 11 and less than 20 are 13, 17, and 19, so Samantha's sum was $12 + 14 + 15 + 16 + 18 = \boxed{75}$.

8. There are $4 \times 4 = 16$ total squares, and after 9 are placed into the recycle bin, $16 - 9 = 7$ remain. Each of these squares measures $8 \div 4 = 2$ inches on a side, and has a perimeter of $4 \times 2 = 8$ inches. The total perimeter of these 7 squares is $7 \times 8 = \boxed{56}$.

9. If Denise cannot be next to Bobby, then at least one of Alice and Carrie must be between Denise and Bobby. But if one of Alice and Carrie is between Denise and Bobby, then both of Alice and Carrie are between Denise and Bobby. Therefore Denise and Bobby must be on either end of the arrangement, and Alice and Carrie must be in the middle. There are 2 ways to place Denise and Bobby on the ends, either Denise on the left end and Bobby on the right, or Denise on the right end and Bobby on the left. Similarly, there are 2 ways to place Alice and Carrie in the middle. The total number of possible arrangements is therefore $2 \times 2 = \boxed{4}$.

10. Because 3 and 5 are both prime, any number that is divisible by both must also be divisible by $3 \times 5 = 15$. The multiples of 15 that are less than 160 are $15 \cdot 1$, $15 \cdot 2$, $15 \cdot 3$, and so on, up to $15 \cdot 10 = 150$. Their sum is $15 \cdot (1 + 2 + 3 + \ldots 10)$, which is $15 \cdot 55 = \boxed{825}$.

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11. As a fraction, $8\frac{1}{3}\%$ is $\frac{1}{100} \cdot \frac{8*3+1}{3} = \frac{1}{12}$. Similarly, 20% is $\frac{1}{100} \cdot \frac{20}{1} = \frac{1}{5}$. The pieces of the pie that were cut totaled $5 \cdot \frac{1}{12} + 2 \cdot \frac{1}{5}$ of the pie, or $\frac{49}{60}$ of the pie. As a fraction of the pie, the final piece was $1 - \frac{49}{60} = \boxed{\frac{11}{60}}$.

12. This is $\frac{3}{2} \div \frac{4}{3} \div \frac{5}{4} \div \frac{6}{5}$, or $\frac{3}{2} \cdot \frac{3}{4} \cdot \frac{4}{5} \cdot \frac{5}{6}$. Canceling yields $\frac{3 \cdot 1 \cdot 1 \cdot 1}{2 \cdot 2}$, which is $\boxed{\frac{3}{4}}$.

13. Because the area is 63, the product of the two leg lengths is $2 \cdot 63 = 126$, so the length of the second leg is $\frac{126}{18} = 7$. By the Pythagorean Theorem, the square of the length of the hypotenuse is $18^2 + 7^2 = 373$. Since $19^2 = 361$ and $20^2 = 400$, the least positive whole number that is greater than the length of the hypotenuse is $\boxed{20}$.

14. There are $23 - 5 = 18$ students in either art or Spanish and $18 - 3 = 15$ students in either art or Spanish only. Let x be the number of students in art but not in Spanish. Then there are $15 - x$ students in Spanish but not art, and $x + 3 = 2 \cdot (15 - x + 3)$. Solving this equation for x, the number of students in art but not in Spanish is $\boxed{11}$.

15. If Pierre has 1 five-point token, he could have $22 \div 2 = 11$ two-point tokens, for a total of $1 + 11 = 12$ tokens. If Pierre has 3 five point tokens, he could have $12 \div 2 = 6$ two-point tokens, for a total of $3 + 6 = 9$ tokens. If Pierre has 5 five-point tokens, he could have $2 \div 2 = 1$ two-point tokens, for a total of $5 + 1 = 6$ tokens. Altogether, the sum of all possible values of the number of tokens is $6 + 9 + 12 = \boxed{27}$.

16. After the pieces are removed, the paper is a hexagon. The remaining edges of the original piece of paper are 4 segments of length 5 inches, for a combined length of $4 \cdot 5 = 20$ inches. The additional 2 edges are hypotenuses of isosceles right triangles with legs of length 5. Each has a length of $5\sqrt{2}$, and their combined length is $10\sqrt{2}$. The total perimeter is $20 + 10\sqrt{2}$ inches, and since $\sqrt{2} \approx 1.41$, the whole number nearest to $20 + 10\sqrt{2}$ is $\boxed{34}$.

17. Let Jamele's number be j. Then $\frac{j}{41} > \frac{1}{5}$, or $5j > 41$. This is true when $j > 8$. Similarly, $\frac{j}{41} < \frac{1}{2}$, or $2j < 41$. This is true when $j < 21$. The number of whole numbers greater than 8 and less than 21 is $\boxed{12}$.

18. Since 1728 is 12^3, the prime factorization of 1728 is $(2^2 \cdot 3)^3 = 2^6 \cdot 3^3$. Any even divisor of 1728 must have an exponent of 2 that is greater than or equal to 1 and less than or equal to 6 and an exponent of 3 that is 0, 1, 2, or 3. The total number of possibilities for these two exponents is $6 \cdot 4 = \boxed{24}$.

19. Consider pairing successive terms in this list and summing the numbers in each pair. The first few terms in the list of these sums are $2 + 1 = 3$, $6 + 5 = 11$, $10 + 9 = 19$, and $14 + 13 = 27$. This is an arithmetic sequence with common difference 8. The sum of the first 20 terms of this sequence is the sum of the first 40 terms of Allen's list. The 20th term of this sequence is $8 \cdot 20 - 5 = 155$, so the sum of the first 20 terms is $\frac{1}{2} \cdot 20 \cdot (3 + 155) = \boxed{1580}$.

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20. The given expression can be rewritten as $(m-n)^3$, so $(3\spadesuit 1) = (3-1)^3$, or 8, and $(2\spadesuit 1) = (2-1)^3$, or 1. Finally, $8\spadesuit 1 = (8-1)^3$, which is $\boxed{343}$.

21. There are $\binom{5}{2} = 10$ ways to choose 2 of the 5 flavors. Each of the three toppings can be either on or off the sunday, so there are $2 \cdot 2 \cdot 2 = 8$ ways to select some, all, or none of the toppings. Altogether, the number of possible sundaes is $10 \cdot 8 = \boxed{80}$.

22. The expression 9^{x-1} is equivalent to $\left(3^2\right)^{x-1}$, or 3^{2x-2}. Since $3^{2x+1} = 54$, the value of 3^{2x-2} is $\frac{54}{3^3} = \boxed{2}$.

23. The distance between the sides of length 9 is $\frac{72}{9} = 8$. Dropping an altitude from a vertex where the longer diagonal meets one side of length 9 forms a right triangle with hypotenuse of length 17 and leg of length 8. By the Pythagorean Theorem, the other leg of this triangle has length $\sqrt{17^2 - 8^2} = 15$. The distance of the foot of the altitude from the nearest vertex is $15 - 9 = 6$. The side of the parallelogram of unknown length is the hypotenuse of a right triangle with legs of length 6 and 8, and has length $\sqrt{6^2 + 8^2} = 10$. The perimeter of the parallelogram is $10 + 10 + 9 + 9 = \boxed{38}$.

24. Every power of 6 requires one factor of 2 and one factor of 3. In the product 120!, factors of 2 will greatly exceed factors of 3, so powers of 6 are constrained by factors of 3. Every multiple of 3 contributes 1 factor of 3, and, discarding the remainder when 120 is divided by 3, there are 40 such numbers in the product. Every multiple of 9 contributes an additional factor of 3, and, discarding the remainder when 120 is divided by 9, there are 13 such numbers in the product. Every multiple of 27 contributes an additional factor of 3, and, discarding the remainder when 120 is dividided by 27, there are 4 such numbers in the product. Finally, $3^4 = 81$ contributes 1 additional factor of 3. Altogether, the number of factors of 3 is $40 + 13 + 4 + 1 = 58$, so the greatest power of 6 that evenly divides 120! is $\boxed{6^{58}}$.

25. There are $2^5 = 32$ possible outcomes. Among these, any outcome with 0 or 5 heads will have 3 or more consecutive results that are the same. If there are only 1 or 4 heads, there is 1 outcome in each case in which 3 or more consecutive results that are the same do not occur, namely, *HHTHH* and *TTHTT*, for a total of 2 outcomes. There are $\binom{5}{2} + \binom{5}{3} = 20$ total outcomes with 2 or 3 heads. Among these, three consecutive identical outcomes can only occur as *HHHTT*, *THHHT*, *TTHHH*, *TTTHH*, *HTTTH*, and *HHTTT*, for a total of 6. Then $20 - 6 = 14$ outcomes do not have three consecutive results that are the same. Altogether, there are $2 + 14 = 16$ outcomes that do not have three consecutive results that are the same, for a probability of $\frac{16}{32} = \boxed{\frac{1}{2}}$.

26. Let the three numbers be $J3$, $K3$, and $L3$, where J, K, and L are digits. Then their product is $(10J + 3)(10K + 3)(10L + 3)$. When expanded, the only terms of this product that impact the tens digit are $90J + 90K + 90L + 27$. Since this value has a tens digit of 3, $90J + 90K + 90L$ must have a tens digit of 1 because $90J + 90K + 90L$ necessarily has a units digit of 0. Dividing by 10, $9 \cdot (J + K + L)$ must have a units digit of 1. This occurs when $J + K + L$ has a units digit of 9. The tens digit of the sum is the units digit of $J + K + L$, since the sum of the numbers is $10 \cdot (J + K + L) + 9$. Since $J + K + L$ has a units digit of 9, the answer is $\boxed{9}$.

27. Let the diameter of the larger circle be x and the diameter of the smaller circle be y. Then $\pi\left(\frac{x}{2}\right)^2 - \pi\left(\frac{y}{2}\right)^2 = 36\pi$, or $x^2 - y^2 = 144$. By difference of squares, $(x+y)(x-y) = 144$. This only has solutions that are positive whole numbers x and y if both $(x+y)$ and $(x-y)$ are even. The factor pairs of 144 with both factors even are $(72, 2)$, $(36, 4)$, $(24, 6)$, $(18, 8)$, and $(12, 12)$. The pair $(12, 12)$ is discarded because that would yield a smaller circle with a diameter of 0. The remaining four pairs yield values of y that are 35, 16, 9, and 5, and their sum is $\boxed{65}$.

28. Including the boundaries of the board, there are $8 + 1 = 9$ possible vertical boundaries and 9 possible horizontal boundaries. A rectangle can be formed by choosing any 2 of the 9 vertical boundaries and any 2 of the 9 horizontal boundaries, for a total of $\binom{9}{2} \cdot \binom{9}{2} = 1296$ possible rectangles. There are $8 \cdot 8 = 64$ squares that are 1×1. For a 2×2 square, there are 7 horizontal possibilities for the lower left corner, and 7 vertical possibilities for the same lower left corner, for a total of $7 \cdot 7 = 49$ squares that are 2×2. For 3×3, the number is $6 \cdot 6 = 36$. This continues, so that the total number of squares is $64 + 49 + 36 + 25 + 16 + 9 + 4 + 1 = 204$. The probability that a randomly chosen rectangle is a square is $\frac{204}{1296} = \frac{17}{108}$, and $17 + 108 = \boxed{125}$.

29. Because $2^{10} = 1024$, all of the positive whole numbers from 1 through $1024 - 1 = 1023$ can be written with 10 or fewer total digits in binary. Considering all of the numbers from 1 through 1023, there are $\binom{10}{6} = 210$ that require exactly 6 ones, $\binom{10}{5} = 252$ that require exactly 5 ones, $\binom{10}{4} = 210$ that require exactly 4 ones, $\binom{10}{3} = 120$ that require exactly 3 ones, $\binom{10}{2} = 45$ that require exactly 2 ones, and $\binom{10}{1} = 10$ that require exactly 1 one. Altogether, the number of positive whole numbers less than or equal to 1023 that require 6 or fewer ones is $210 + 252 + 210 + 120 + 45 + 10 = 847$. But this includes numbers greater than 999. The binary number 1111100000_2 is equal to $512+256+128+64+32 = 992$. If the a sixth one appears in the place with value 16 or 8, then the value will exceed 999. Therefore, the number of positive whole numbers less than or equal to 999 that require 6 or fewer ones when written in binary is $847 - 2 = \boxed{845}$.

30. Triangle BPQ is similar to triangle DPA, so $\frac{36}{AP} = \frac{BP}{PD}$. Additionally, triangle BPA is similar to triangle DPR, so $\frac{AP}{36+28} = \frac{BP}{PD}$. Therefore $\frac{36}{AP} = \frac{AP}{36+28}$, so $AP^2 = 36 \cdot (36 + 28)$, or $36 \cdot 64$. Therefore the length of segment AP is $\sqrt{36 \cdot 64} = \boxed{48}$.

Target Round Solutions

1. One way to solve this is by working backwards. Subtract 5 from the number, then 6 from that result, then 7 from that result, and so on, until the result is 0. Taking this approach, the first few steps are $200 - 5 = 195$, $195 - 6 = 189$, $189 - 7 = 182$, $182 - 8 = 174$. The final few steps are $57 - 18 = 39$, $39 - 19 = 20$, and $20 - 20 = 0$. Therefore, the number Trey chose was $\boxed{20}$.

2. The entire figure is comprised of 4 triangles that do not overlap. There is also the 1 triangle that is the entire figure. Finally, there are 2 triangles that have a vertex at the lower left vertex of the figure and a third side that is one of the 3 segments internal to the figure. Altogether, the number of triangles in the figure is $4 + 1 + 2 = \boxed{7}$.

3. Since $31 \times 31 = 961$ and $32 \times 32 = 1024$, there are 31 perfect squares less than 1000. The units digit can only be odd if the units digit of the number being squared is also odd. The number of positive odd numbers less than or equal to 31 is $\boxed{16}$.

4. When the tank is 40% full, it is 60% empty, so $500 \cdot 0.6 = 300$ gallons of water are needed to fill the tank. In the first 16 minutes, $16 \cdot 2\frac{1}{2} = 40$ gallons of water are added to the tank, so $300 - 40 = 260$ gallons remain to be added. From this point on, water is added to the tank at a rate of $2\frac{1}{2} - \frac{1}{3} = \frac{13}{6}$ gallons per minute. This will require $260 \div \frac{13}{6} = 120$ minutes, so the total number of minutes required for the tank to be full is $16 + 120 = \boxed{136}$.

5. Without the 2 extra dimes, Dylans coins are worth $155 - 2 \cdot 10 = 135$ cents. These coins can be grouped into pairs of 1 nickel and 1 dime, with each group worth $5 + 10 = 15$ cents. There are $135 \div 15 = 9$ such pairs, comprised of $2 \cdot 9 = 18$ coins. Including the 2 extra dimes, the total number of coins is $18 + 2 = \boxed{20}$.

6. There are 3 distinct cases where the product is not a multiple of 4. In the first case, all 3 rolls are odd. This occurs with probability $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$. In the second case, 2 of the rolls are odd, and one of the rolls is a 2. There are 3 choices for the die that shows a 2, for a probability of $3 \cdot \frac{1}{6} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$. In the third case, 2 of the rolls are odd, and one of the rolls is a 6. This is equivalent to the second case, and also has probability $\frac{1}{8}$. The total probability of the product not being a multiple of 4 is $\frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8}$, so the probability of the product being a multiple of 4 is $1 - \frac{3}{8} = \boxed{\frac{5}{8}}$.

7. April 1st is $31 + 28 + 31 = 90$ days after January 1st. Since $91 = 13 \times 7$, this means that April 1st is one day before Friday, or Thursday. Then April 2nd, 9th, 16th, 23rd, and 30th are Fridays, and April 3rd, 10th, 17th, and 24th are Saturdays, for a total of Fridays and Saturdays of $5 + 4 = \boxed{9}$.

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8. Consider the trapezoid formed by the centers of the circles of radii 30 and 45 and the points where these circles are tangent to the line. By the Pythagorean Theorem, the altitude of this trapezoid is $\sqrt{(45+30)^2-(45-30)^2} = 30\sqrt{6}$. This trapezoid has area $\frac{1}{2}\cdot(45+30)\cdot 30\sqrt{6} = 1125\sqrt{6}$. Now consider the trapezoid formed by the centers of the circles of radii 45 and 60 and the points where these circles are tangent to the line. The altitude of this trapezoid is $\sqrt{(60+45)^2-(60-45)^2} = 60\sqrt{3}$, and the area of this trapezoid is $\frac{1}{2}\cdot(45+60)\cdot 60\sqrt{3} = 3150\sqrt{3}$. Finally, consider the trapezoid formed by the centers of the circles of radii 30 and 60 and the points where these circles are tangent to the line. This trapezoid has area $\frac{1}{2}\cdot(30+60)\cdot(30\sqrt{6}+60\sqrt{3}) = 1350\sqrt{6}+2700\sqrt{3}$. The area of the triangle formed by the centers of the circles is the sum of the areas of the first two trapezoids minus the area of the third, or $1125\sqrt{6}+3150\sqrt{3}-1350\sqrt{6}-2700\sqrt{3} = 450\sqrt{3}-225\sqrt{6}$. To the nearest whole number, this is $\boxed{228}$.

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Team Round Solutions

1. There are 60 minutes in 1 hour, so if the episode was 60 minutes long, it would have concluded at 1:36 pm. But it concluded 9 minutes before this time, so the length of the episode, in minutes, was $60 - 9 = \boxed{51}$.

2. For the value of the number to increase by 45 when the digits are reversed, the units digit must exceed the tens digit by $45 \div 9 = 5$. The two-digit numbers with this property are 16, 27, 38, and 49, so altogether there are $\boxed{4}$.

3. The sum of the ages of the 6 people on the trip was $6 \times 26 = 156$. When the 11 year old and the 37 year old left, the sum of the ages of the remaining $6 - 2 = 4$ people on the trip was $156 - 11 - 38 = 108$, and their average age was $108 \div 4 = \boxed{27}$.

4. The absolute speed each travels at does not matter. Since they have both traveled for the same amount of time, Beth has traveled 4 times as far as Steph. Therefore, Beth has traveled $\frac{4}{4+1} = \frac{4}{5}$ of the total distance, while Steph has traveled $\frac{1}{4+1} = \frac{1}{5}$ of the total distance. The difference between the two distances each travels is $\frac{4}{5} - \frac{1}{5} = \frac{3}{5}$ of 6 miles, or $\frac{18}{5}$ miles. As a mixed number, this is $\boxed{3\frac{3}{5}}$.

5. The number in the top left must produce the same sum with its column, 21 and 22, as with its row, which is 17 and the number in the top right. Then number in the top right must be $21 + 22 - 17 = 26$. The number in the center square is common to the diagonal that includes the top right and the middle column. Using the same reasoning as before, the number in the bottom center square must be $22 + 26 - 17 = 31$. The bottom row is then complete, and has a sum of $22 + 31 + 19 = 72$. The top left square is $72 - 21 - 23 = 29$, the center square is $72 - 17 - 31 = 24$, and the middle right square is $72 - 26 - 19 = 27$. The sum of the numbers that belong in the five empty smaller squares is $26 + 31 + 29 + 24 + 27 = \boxed{137}$.

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6. Let the measure of the angle be x. Then $x = \frac{1}{2}(90^\circ - x) - 9^\circ$. Rearranging and multiplying by 2 yields $2x + 18^\circ = 90^\circ - x$, so $3x = 72^\circ$, and $x = \boxed{24^\circ}$.

7. Suppose the smallest whole number chosen was 2. Then the possible collections of three numbers are $(2, 2, 18)$, $(2, 3, 12)$, $(2, 4, 9)$, and $(2, 6, 6)$. The first and last can be permuted among the three in 3 ways, while the other three can be permuted among the three in $3! = 6$ ways. When the smallest number is 2, there are a total of $2 \cdot 3 + 2 \cdot 6 = 18$. If the smallest number chosen was 3, the only possible collections of three numbers are $(3, 3, 8)$ and $(3, 4, 6)$. The first possibility can be permuted among the three in 3 ways, while the second can be permuted among the three in 6 ways. Altogether, the total number of possibilities is $18 + 3 + 6 = \boxed{27}$.

8. Let the price of the half-price tickets be p and the number of full-price tickets be x. Then the number of half-price tickets is $80 - x$, the price of the full-price tickets is $2p$, and $2p \cdot x + p \cdot (80 - x) = 1122$. Solving this for p yields $p = \frac{1122}{x+80}$. But p must be a whole number, and x must be a positive whole number less than 80. Therefore $x + 80$ must be a divisor of 1122 that is between 80 and 160. The prime factorization of 1122 is $2 \cdot 3 \cdot 11 \cdot 17$, so the only possible value of $x + 80$ is $2 \cdot 3 \cdot 17 = 102$. The number of full-price tickets is $102 - 80 = 22$, and the number of half-price tickets is $80 - 22 = \boxed{58}$.

9. Let the radius of the larger circle be 6. Then the area of the larger circle is $6^2 \cdot \pi = 36\pi$. The side length of equilateral triangle BCX is also 6. The altitudes of BCX have length $\frac{1}{2} \cdot 6 \cdot \sqrt{3} = 3\sqrt{3}$. In an equilateral triangle, altitudes are medians, and the centroid, or point where the medians intersect, is also the center of the circumscribed circle. The distance from the centroid to a vertex of the triangle is the radius of the circumscribed circle, and is $\frac{2}{3}$ of the altitude, or $2\sqrt{3}$. Therefore the area of the smaller circle is $(2\sqrt{3})^2 \cdot \pi = 12\pi$, and the ratio of the area of the smaller circle to the area of the larger circle is $\frac{12\pi}{36\pi} = \boxed{\frac{1}{3}}$.

10. Suppose the ant can move through the interior of the large cube. The journey from A to B would be comprised of two moves up, two moves right, and two moves back. The number of ways to arrange these moves is the number of permutations of $UURRBB$, which is $\frac{6!}{2! \cdot 2! \cdot 2!} = 90$. However, the ant cannot visit the center of the $2 \times 2 \times 2$ cube. This only occurs on any path where the first three moves are some permution of URB, and the last three moves are also some permutation of URB. This occurs in $3! \cdot 3! = 36$ ways. The total number of paths the ant can take from A to B is therefore $90 - 36 = \boxed{54}$.

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